Tutorial 2 – Fundamental knowledge

T2-1. Calculate the concentration in $[mg/L]$ of a solution with a density of $\rho = n\ g/mL$ and a concentration of 1 ppm (diluted solution).

T2-2. How many mL of concentered HCl are needed to make a 500 mL solution with a concentration of 0.250 M HCl? The concentrated HCl is 37.2% (w/w) and has a density of 1.188 g/cm^3. (MW_{HCl} = 36.46 g/mol)

T2-3. A limiting reagent in a chemical reaction is the reagent that is consumed first. Once the limiting reagent is gone, the reaction ceases. To decide which reagent is the limiting one:

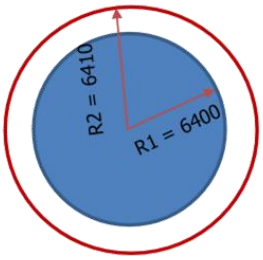
a. Find the number of moles of each available reactant.

b. Compare the number of moles present to the number of moles required for a complete reaction.

c. Determine the limiting reagent, which is less moles provided or produce less product.

Determine the limiting reagent in a reaction of 1.00 g $CaCl_2$ (110.98 g/mol) with 1.15 g $Na_2C_2O_4$ (134.00 g/mol) in water. Calculate the unreacted reagent fraction (in %)?

T2-4. Check the validity of the statement "In every single breath we take, we inhale at least one molecule from the dying breath of Julius Caesar". Make the following assumptions:



a. Air particles are distributed evenly all over the atmosphere.

b. No particles are created or destroyed.

c. The earth radius is 6400 km.

d. The atmosphere height is 10 km.

e. The average volume of air in a single breath is 0.5 L.

f. The air follows the ideal gases law (PV = nRT, R = 0.082 $\frac{\text{L}\times\text{atm}}{\text{mol}\times\text{K}}$ = 0.08314 $\frac{\text{L}\times\text{bar}}{\text{mol}\times\text{K}}$).

g. The earth is under standard conditions (T = 298 K, P = 1 atm).